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Data-Driven Volatile Cryptocurrency Price Forecasting via Variational Mode Decomposition and BiLSTM

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Cryptocurrency is based on blockchain technology which is ideally decentralised, referring to no superior authority overlooking it. The community is maintained by numerous user machines forming a “peer-to-peer” network. With the recent skyrocket of crypto-assets in the financial markets, many view it as the quickest and riskiest way to earn. Such assets are coined as “Volatile” due to its rapidly fluctuating price, thereby making it extremely hard to forecast its course. The paper at hand explores a novel technique that establishes a relation between signal processing and volatile stock forecasting methods via variational mode decomposition (VMD). Variational mode decomposition aided with BiLSTM neural architecture, a purely data-driven model, is fine-tuned to forecast the daily or interday prices of Bitcoin and Ethereum alongside yielded RMSE of 0.0278 for Bitcoin. The results are then further compared with ARIMA, ARMA and MA to analyse the effect of VMD.

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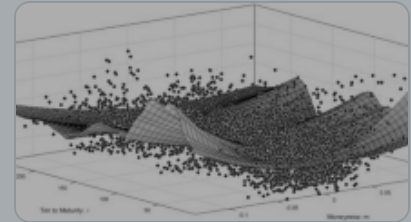
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